

First name:

Last name:

Student ID:

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Problem 1. Use a triple integral, thanks to the Divergence Theorem, to evaluate the surface integral $\iint \vec{F} \cdot \vec{n} \, dA$ for

$$\vec{F} = [x^2, 0, z^2],$$

S is the surface of the box $|x| \leq 1$, $|y| \leq 3$, $0 \leq z \leq 2$.

[5 marks]

Solution.

□

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Problem 2. *Verify Stokes' Theorem for*

$$\vec{F} = [y^3, -x^3, 0],$$

and S is the surface

$$x^2 + y^2 \leq 1, \quad z = 0.$$

Hint: To save time, you may directly use, if needed, the following facts:

- 1. S can be parameterized as $x = u \cos v$, $y = u \sin v$, for $0 \leq u \leq 1$, $0 \leq v \leq 2\pi$, and in this set up, $\vec{N} = \vec{r}_u \times \vec{r}_v = u \hat{k}$.*
- 2. The boundary ∂S , can be parameterized as $x(t) = \cos(t)$, $y(t) = \sin(t)$, $0 \leq t \leq 2\pi$.*
- 3. The expression $\sin^4(t) + \cos^4(t)$ can be simplified to $\frac{3}{4} + \frac{1}{4} \cos(4t)$.*

[5 marks]

Solution.

□

MATH73235

Spring 2025

Quiz 5

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